

Predicting Cas9 gene-editor activity: proposal evaluation, architecture and experiment

1. What the measurements support, and the dataset's most consequential limitations

CAS9_STRP1_Spencer_2017_positive reports, for 8,117 single amino-acid substitutions in *S. pyogenes* Cas9 (4-7 of the 19 possible substitutions per position, 1,368 positions, 100% single-mutants), a log2 fold-change in variant frequency after one round of positive bacterial selection for on-target nuclease activity (survival requires expression + correct cleavage of a single fixed target). This supports predicting **relative, assay-specific nuclease-competence** of single substitutions on one target/gRNA in *E. coli* — not mammalian editing efficiency, not specificity, not multi-mutant epistasis. The original authors validated a subset in mammalian cells with good but imperfect concordance (21/25 single mutants agreed in direction).

Four limitations most affect any predictive claim:

1. Library-construction artefact. The library was made by error-prone PCR

(0.18% substitution rate), so it can only sample amino-acid changes reachable by a *single* nucleotide substitution. We verified this directly: 100% of the 8,117 measured substitutions are single-nucleotide reachable, versus only **39.5%** of all 380 possible wt→mut amino-acid pairs (diagnostics.json, frac_all_aa_pairs_single_nt_reachable). The assay therefore never measures the ~60% of substitutions (including most physicochemically drastic ones) that require ≥2 nucleotide changes — a systematic, non-random gap in mutation space that any generalization claim must respect.

2. Coarse, one-sided readout. Scores are heavy-tailed (excess kurtosis 10.2,

range -6.45 to +3.92) and left-skewed: most substitutions are neutral-to-mildly deleterious (median -0.27) and few are strongly beneficial. The provided binary label is a **median split** of this same continuous score — not a biologically grounded activity threshold — so treating DMS_score_bin as ground truth for "functional/non-functional" is unjustified without independent validation.

3. Sparse, non-saturating coverage. Coverage is 0.31× of the full 19×L

substitution space, all from ep-PCR's biased 39.5%-reachable subset — this is deep mutational scanning in name but not a saturation assay.

4. Position is the dominant signal, but only observed, not causally isolated.

Fitting one mean score per position accounts for $R^2=0.221$ of variance in-sample and Spearman ≈ 0.16 leave-one-out (spearman_LOO_position_mean); a ±15-residue neighbour-position average gives the same ≈ 0.15 -0.16. So most predictable variance is a smooth positional (domain/structural) tolerance signal, not amino-acid-substitution chemistry — any evaluation that lets a model see same- or nearby-position labels at train time will over-credit models for memorizing this positional average rather than learning transferable substitution or sequence-context effects. (We found no informative missingness: variants-per-position does not correlate with position-mean score, Spearman 0.033, ns — coverage gaps are not concentrated at extreme positions.)

2. Evaluation protocol

The generalization claim under test is: **"can a model, given no activity labels from the target position or its immediate neighbourhood, rank substitution effects at that position?"** — the realistic use case (scoring a not-yet-assayed engineering target). We therefore run four nested CV protocols (cas9lib.make_splits, train_mask), 10 folds each, and report **position-cluster bootstrap 95% CIs** (resample positions with replacement, 300 resamples) rather than i.i.d. CIs, since rows from the same position are not exchangeable:

- **random** (variant-level 10-fold) — positive control; same-position labels leak. - **unseen position** (position-level 10-fold, no buffer) — adjacent positions can still co-occur across train/test. - **unseen block** (10 contiguous ~137-aa blocks + all training variants within 15 residues of a held-out block removed) — the primary protocol. - **unseen domain** (leave-one-domain-out over RuvC/REC/HNH/PI, +15-residue buffer) — hardest, most decision-relevant split (engineering a domain with no local data).

Baselines: BLOSUM62 substitution score (no fitting); non-learned position/mutant-aa lookups; a codon-reachability-only model (isolates the library artefact). Metrics: pooled Spearman (headline), fold-mean±sd (stability), AUROC on the provided binary label, **within-position Spearman** (ranking substitutions at the same held-out position — the metric closest to actual engineering use, since cross-position variance can inflate pooled Spearman), and top-decile enrichment.

3. Predictor, extension, ablation, and a stress test against my own reading

M1 (implemented predictor of the supplied scores): substitution-level gradient-boosted trees (HistGradientBoostingRegressor) on wt/mut one-hot identity, BLOSUM62 score, and 6 physicochemical property deltas (Kyte-Doolittle hydrophathy, volume, charge, Grantham polarity, Chou-Fasman α/β propensity) — answers "does this substitution chemistry matter, independent of position."

E (substantive extension, two-stage): a *position-tolerance* model predicts each position's mean winsorised score from a **local WT-sequence-context window** around that position only (±2/5/10/20-residue running means of the same six properties + composition flags, fit with buffered inner-CV so it never sees labels near its own test positions) — this stage is what a "trunk producing a position-level embedding" would need to contribute if it worked. Its prediction is then interacted with M1 (E_twostage; M2_sub_ctx is the version without the two-stage structure, i.e. context features concatenated directly). **This tests whether locally-computed sequence context is transferable across positions** — the one part of the original proposal (a context/structure trunk feeding an activity head) that is testable without a pretrained encoder.

Held-out predictions for every (variant, model, protocol) are in predictions.csv; splits in split_assignments.csv.

Ablation (unseen-block split, Fig. 1b): stripping M1 to BLOSUM62 alone (0.112 [0.092,0.133]) or physicochemistry alone (0.129 [0.108,0.152]) costs almost nothing versus the full M1 (0.128 [0.106,0.151]) — one-hot identity adds no measurable signal beyond chemistry, meaning BLOSUM62 is already close to a sufficient statistic here. Codon-reachability alone is weak but non-null (0.032 [0.012,0.055]) — transitions are on average milder substitutions than transversions, so the *library artefact correlates with, but does not add independent signal beyond, real biochemical similarity* (identity+reachability 0.131 [0.110,0.153] $\approx$ M1 alone). Adding context (M2_sub_ctx, 0.068 [0.045,0.092]) **underperforms M1 alone**, and shuffling the context columns across positions barely changes it (M2_shufctx 0.101 [0.075,0.127]) — the 9-fold-larger context feature block is mostly adding noise/overfitting capacity to the GBM, not real signal (E_shufctx 0.119 [0.097,0.141] vs. E_twostage 0.118 [0.095,0.141]: statistically indistinguishable, Δ Spearman 95% CI [-0.020, -0.001] — a wash). Labels-shuffled controls (M2_shuffLeY, E_shuffLeY) correctly collapse to ≤ 0 .

Stage-1 site model, and the stress test. My working hypothesis going in was that local sequence composition around a position predicts its mutational tolerance (a stand-in for "a structure/context trunk should help"). This failed: the GBM site model scores **-0.011 [-0.037, 0.014]** pooled Spearman on unseen blocks and **-0.016 [-0.041, 0.009]** on unseen domains — indistinguishable from zero, and it overfit its own inner-CV (a post-hoc ridge-regression version, disclosed as such in Fig. 1b/README, does better but still only 0.036-0.062, well below the 0.16 LOO position-mean *oracle* that has actual labels). This is evidence *against* my own preferred interpretation: **local WT-sequence composition, as engineered here, is a weak-to-absent predictor of a position's measured mutational tolerance out-of-domain**. Whatever drives the strong in-sample position effect ($R^2=0.22$) is not well captured by a small window of hydrophathy/volume/charge/secondary-structure-propensity statistics — it is more likely true 3-D structural context (contacts, burial, catalytic-site proximity)

that this sandbox could not compute (§4).

Fig. 1 shows the full picture: under **random/posgroup** splits, naive position-mean lookups (local_mean, posmean, 0.155-0.167) are *competitive with or better than* every fitted model — exactly the position-leakage failure mode predicted in §2 — while under the buffered **block/domain** splits every model compresses toward BLOSUM62 (~0.09-0.13), and AUROC on the median-split label drops to 0.53-0.56 (block) / 0.52-0.54 (domain), barely above chance. Within-position Spearman for M1 (0.139 block, 0.128 domain) tracks its pooled Spearman closely, indicating the signal M1 does have is genuine substitution-level ranking ability, not an artefact of cross-position variance.

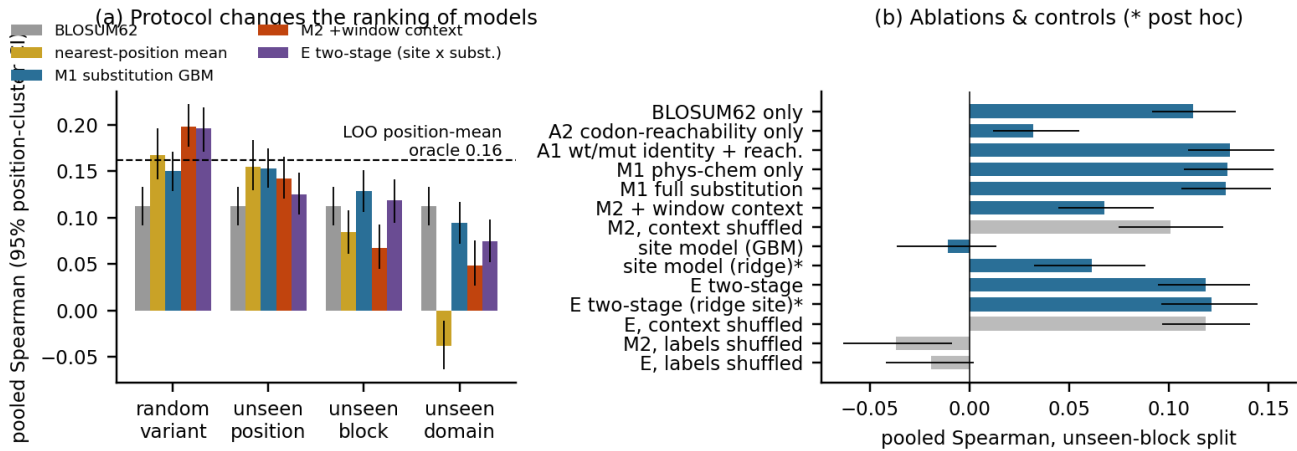


Fig. 1. (a) Protocol determines which model looks best: naive position-mean

lookup baselines rival fitted models under random/posgroup splits, but the advantage disappears once buffered blocks or whole domains are held out — the generalization claim only survives (b) under the honest protocol. (b) Ablations: context features add no more than noise beyond substitution identity; the codon-reachability library artefact is not independently informative; the site/context model's negative-to-near-zero score is the stress test against my own hypothesis. Error bars: 95% position-cluster bootstrap CI, n=300.*

4. Proposed architecture (specification, not implemented here)

Sequence-representation module (retain, frozen): ESM2 esm2_t33_650M_UR50D (pretrained masked protein language model). Interface: full 1,368-aa WT and each mutant sequence → final-layer per-residue embeddings, shape ($L=1368$, $d=1280$); additionally compute **zero-shot masked-marginal scores** $\Delta = \log p(\text{mut_aa} \mid \text{context, pos masked}) - \log p(\text{wt_aa} \mid \text{context, pos masked})$ per variant (a scalar), which is the standard, validated way to get activity- correlated signal from a frozen PLM without any fine-tuning. Optionally add a retrieval-augmented independent-site or EVE-style log-odds ratio from the already-available MSA (CAS9_STRP1_2023-08-07_b01.a2m, $N_{\text{eff}}/L=1.14$ "Medium" per the supplied metadata) as a second frozen zero-shot feature — orthogonal evolutionary information the PLM alone may miss for a prokaryotic, moderately-diverged family.

Structure module — replace the diffusion trunk, don't reuse it as-is. The proposal's "diffusion structure-prediction trunk" (e.g. an RFDiffusion/ESM3-style denoiser) is trained to iteratively remove noise from 3-D coordinates/frames conditioned on a diffusion timestep; it has no native interface that emits a single static, timestep-independent embedding suitable for a downstream scalar regression head — using it would require inventing and separately validating a timestep-selection or noise-averaging scheme, which is out of scope and unmotivated by anything in this assignment. If structural information is wanted, the appropriate substitute is a **non-generative** structure encoder over the single available crystal structure (PDB 5F9R, referenced in metadata but not fetchable in this sandbox): e.g. ESM-IF1 or a GVP-GNN, retained/frozen, producing per-residue solvent accessibility / contact-count / catalytic-site-distance features, shape (L , k) for small k (≈ 5 -10 hand-interpretable structural scalars is sufficient and more auditable than a learned embedding here, given $n=8k$ labels).

Activity head (train, the only trainable part): small GBM or 2-layer MLP over [ESM2 masked-marginal Δ , MSA log-odds Δ (optional), PLM embedding cosine similarity(mut,wt), structural scalars at the mutated position, BLOSUM62, physicochemical deltas] → scalar DMS_score prediction. **Training objective:** primarily a pairwise/margin ranking loss (or Spearman-surrogate), since the deliverable is a within-target *ranking* of variants, with MSE as an auxiliary term for calibration; trained and selected under the same position-clustered, 15-residue-buffered CV as §2, never i.i.d. CV.

Retain/replace/freeze/train summary: retain ESM2 and (optionally) the MSA model as frozen zero-shot scorers — their pretraining objective (masked-token recovery over hundreds of millions of natural sequences) is a good match for "does this substitution look tolerable," and fine-tuning 650M parameters on 8,117 labels clustered in ~1,368 positions would almost certainly overfit the positional structure documented in §1, especially with no GPU/time budget to validate that risk here. Replace the diffusion trunk with a lightweight non-generative structure encoder, frozen, for the reason above. Train only the small head. This whole architecture is a **specification** distinguished from the executed experiment in §3: nothing here about PLM embeddings or structural features has been run — the sandbox has no reachable GPU/model host (huggingface.co, dl.fbaipublicfiles.com, files.rcsb.org all return 403 host_not_allowed) — so §3's hand-crafted-feature experiment is the closest testable proxy for "does local context/position information, however obtained, generalize," and it answered *no* for the cheap version of that question.

5. Recommendation

Modify, don't pursue as literally specified, and don't reject outright. The literal proposal — splice a diffusion structure trunk into an activity predictor — should not be pursued as written: (a) diffusion trunks have no natural static- embedding interface for a regression head (§4); (b) this dataset's own strongest signal (positional tolerance, $R^2=0.22$ in-sample) collapsed to Spearman ≈ 0 under my out-of-domain stress test using cheap sequence-context features, so there is no evidence yet that any context/structure signal generalizes here, let alone one worth a full diffusion trunk; (c) the dataset's substitution-space is a biased ~40%-reachable sample and its binary label is an arbitrary median split, so conclusions from it should stay modest. What the evidence *does* support: a frozen pretrained-PLM zero-shot score plus a small trained head is a well-motivated, cheap, buildable next step, evaluated only under position-buffered CV.

What this experiment establishes: under honest (buffered block/domain) held-out evaluation, substitution chemistry (BLOSUM62/physicochemistry) carries real, reproducible signal (pooled Spearman ≈ 0.09 -0.13, within-position Spearman ≈ 0.13 -0.14, both well above the shuffled-label floor of ≈ 0); naive

position-mean lookups look artificially strong only when evaluation leaks nearby-position labels; and hand-crafted local sequence context adds no detectable signal beyond identity/chemistry once evaluated honestly.

What remains untested: whether a real pretrained PLM's masked-marginal score or embeddings (not the hand-crafted proxy used here) carries transferable position-level signal that this cheap context model failed to find; whether actual structural features (burial, catalytic-site distance from 5F9R) raise the ceiling toward or past the 0.16 LOO position-mean oracle; and whether any of this transfers to the thing ultimately of interest — mammalian on-target/off-target editing efficiency, which this bacterial assay only partially proxies (§1).

Next experiment that could change this recommendation: compute ESM2 masked-marginal scores and 5F9R-derived burial/contact features for all 8,117 variants (both blocked here only by network access, not by any conceptual gap), and rerun exactly the four-protocol, position-cluster-bootstrap evaluation already built in this repository (`run_experiments.py`) with those as additional feature columns. If PLM/structural features beat the M1 chemistry-only baseline under the buffered block/domain splits by a margin outside the bootstrap CI, that would justify building the full head-only architecture in §4; if they don't, it would justify simplifying further, to chemistry-only substitution scoring.